

Module “4”: Assignment 1 Answer Key

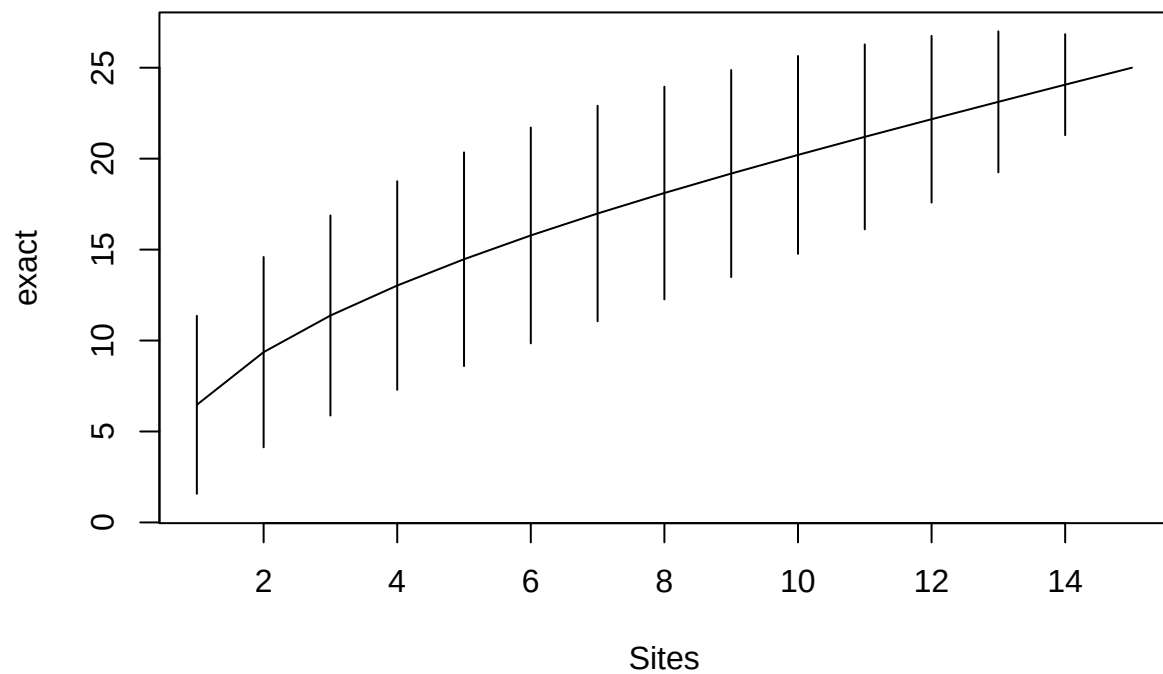
Ellen Bledsoe

2026-05-04

4.

```
## Warning in cor(x > 0): the standard deviation is zero
## Species Accumulation Curve
## Accumulation method: exact
## Call: specaccum(comm = scr)
##
##
## Sites      1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000
## Richness 6.466667 9.361905 11.375824 13.024176 14.470529 15.778821 16.984149
## sd       2.445858 2.615566 2.749919 2.866508 2.936565 2.966045 2.962647
##
## Sites      8.000000 9.000000 10.000000 11.000000 12.000000 13.000000 14.000000
## Richness 18.111422 19.180020 20.205128 21.198535 22.169231 23.123810 24.066667
## sd       2.923536 2.845109 2.719706 2.542279 2.290225 1.937607 1.388844
##
## Sites      15
## Richness 25
## sd        0
```

5.



7.

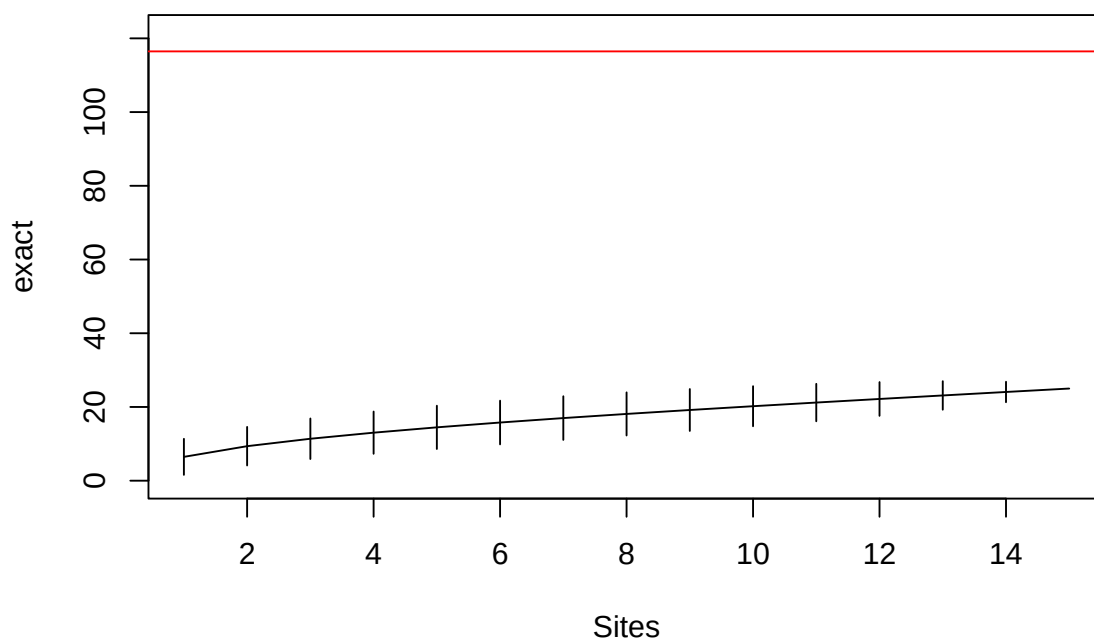
```
## Species      chao  chao.se  jack1 jack1.se  jack2      boot boot.se  n
## All         25 116.4667 104.1535 38.06667 6.191033 49.39524 30.15487 2.9969 15
```

8.

```
## [1] 25
## [1] 116.4667
```

9.

```
## [1] 91.46667
```



10.

11.

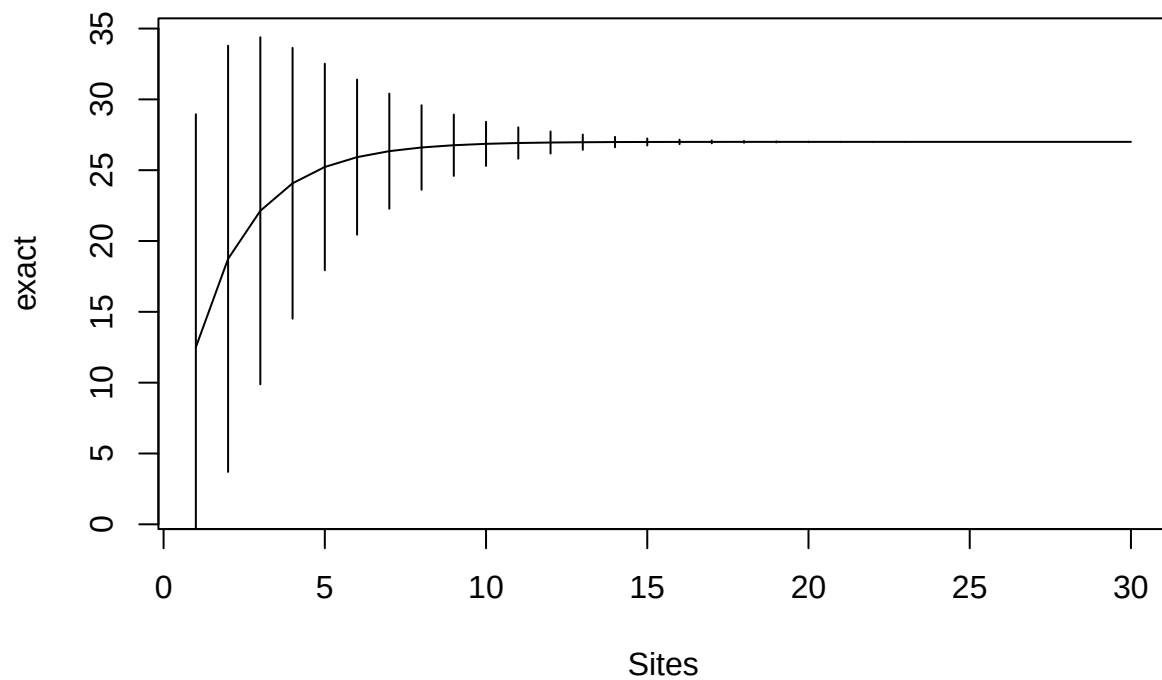
```
##   CHA TRU VAI LOC OMB BLA HOT TOX VAN CHE BAR SPI GOU BRO PER BOU PSO ROT CAR
## 1   0   3   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
## 2   0   5   4   3   0   0   0   0   0   0   0   0   0   0   0   0   0   0
## 3   0   5   5   5   0   0   0   0   0   0   0   0   1   0   0   0   0   0
## 4   0   4   5   5   0   0   0   0   0   1   0   0   1   2   2   0   0   0
## 5   0   2   3   2   0   0   0   0   5   2   0   0   2   4   4   0   0   2
## 6   0   3   4   5   0   0   0   0   1   2   0   0   1   1   1   0   0   0
##   TAN BCO PCH GRE GAR BBO ABL ANG
## 1   0   0   0   0   0   0   0   0
## 2   0   0   0   0   0   0   0   0
## 3   0   0   0   0   0   0   0   0
## 4   1   0   0   0   0   0   0   0
## 5   3   0   0   0   5   0   0   0
## 6   2   0   0   0   1   0   0   0
```

14.

```
## Species Accumulation Curve
## Accumulation method: exact
## Call: specaccum(comm = fish)
##
##
## Sites      1.000000  2.000000  3.000000  4.000000  5.000000  6.000000  7.000000
## Richness 12.500000 18.742529 22.133498 24.076665 25.226910 25.921113 26.344608
## sd       8.224962  7.521837  6.125741  4.780134  3.645886  2.738364  2.032045
##
## Sites      8.000000  9.000000 10.000000 11.000000 12.000000 13.000000 14.000000
```

```
## Richness 26.604116 26.763081 26.860035 26.918697 26.953776 26.974418 26.986307
## sd       1.491678 1.083656 0.778985 0.553784 0.388989 0.269569 0.183966
##
## Sites    15.000000 16.00000 17.000000 18.000000 19.000000 20.000000 21.000000
## Richness 26.992969 26.99657 26.998432 26.999339 26.999749 26.999917 26.999978
## sd       0.123247 0.08070 0.051322 0.031424 0.018253 0.009885 0.004868
##
## Sites    22.000000 2.30e+01 24 25 26 27 28 29 30
## Richness 26.999996 2.70e+01 27 27 27 27 27 27 27
## sd       0.002031 7.01e-04 0 0 0 0 0 0 0
```

15.



16.

```
## Species chao chao.se jack1 jack1.se jack2 boot boot.se n
## All 27 27 0 27 0 27 27.00065 0.02849795 30
```

17.

```
## [1] 27
```

```
## [1] 27
```

18.

```
## [1] 0
```

19.

